

2024 AMC 10A Problem 11 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 11. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 11

Problem:

How many ordered pairs of integers (m, n) satisfy $\sqrt{n^2 - 49} = m$?

Answer Choices:

- A. 1
- B. 2
- C. 3
- D. 4
- E. infinitely many

Solution:

Notice that $m \geq 0$, and if (m, n) is a solution, then so is $(m, -n)$. Assume $n \geq 0$. Squaring both sides of the given equation gives $n^2 - 49 = m^2$, so $n^2 - m^2 = (n - m)(n + m) = 49$. Because $n - m$ and $n + m$ are positive integers, either $n - m = n + m = 7$, or $n - m = 1$ and $n + m = 49$. The first case gives $(0, 7)$ as a solution, and the second case gives $(24, 25)$ as a solution. Because n is squared in the given equation, the corresponding negative values for n also give solutions: $(0, -7)$ and $(24, -25)$. There are (D) 4 solutions in all.

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